



Lab Number: 02

Course Code: CL2001

Course Title: Data Structures - Lab

Semester: Fall 2026

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Guidelines

- You may use **only the concepts covered in the lab** to complete this task.
- Your submission must be a **single PDF file** that includes:
 - The source code (in plain text)
 - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
Example: *25p-1234_ozair_labtask02.pdf*
- Unethical use of AI tools will result in **0 marks**.
- Late submissions will incur a penalty of **-3 marks per day**.

Tasks

ONLY USE POINTER NOTATION TO IMPLEMENT ALL THE SOLUTIONS. ARRAY INDEXING [] IS NOT ALLOWED.

1 Leaders of an Array

An element in an array is called a **leader** if it is greater than every element that appears to its right.

Given an array of n integers, write a C++ program to find all the **leaders** in the array.

The leaders must be stored in a **new dynamically allocated array** using dynamic memory allocation (`new`).

The last element of the array is always considered a leader because there are no elements to its right.

Input:

- The first line contains an integer n , representing the number of elements in the array.
- The second line contains n integers representing the elements of the array.

Output:

- Display all the leader elements in their original left-to-right order.
- The leaders must be stored in a newly dynamically allocated array.

Example:

Input:

```
6
16 17 4 3 5 2
```

Output:

```
17 5 2
```



2 Dynamic Memory Calculator

Create a program that dynamically allocates two arrays of integers of size n. The program should calculate:

1. Sum of corresponding elements
2. Difference of corresponding elements
3. Product of corresponding elements

Store each result in a separate dynamically allocated array.

Example Input:

A: 2 4 6
B: 1 3 5

Example Output:

Sum: 3 7 11
Difference: 1 1 1
Product: 2 12 30

3 Two Sum

Given an integer array `nums` and an integer `target`, find the indices of the two elements whose sum equals `target`.

You may assume that each input would have exactly one solution, and you may not use the same element twice.

Define the following function:

```
int* findIndices(int* nums, int target);
```

The function must:

1. Dynamically allocate an integer array of size 2.
2. Store the two indices in the array.
3. Return the dynamically allocated array.

Examples:

Input: `nums = [2,7,11,15], target = 9`
Output: `[0,1]`

Input: `nums = [3,2,4], target = 6`
Output: `[1,2]`

Input: `nums = [3,3], target = 6`
Output: `[0,1]`



3 Find the Single Non-Repeating Element

Write a C++ program that, given a non-empty array of integers in which every element appears exactly twice except for one element, finds and returns the element that appears only once.

Examples:

- Input: [2, 2, 1] : Output: 1
- Input: [4, 1, 2, 1, 2] : Output: 4
- Input: [1] : Output: 1

4 Roman Numeral to Integer Conversion

Roman numerals are represented by seven different symbols: I, V, X, L, C, D and M.

Symbol	Value
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

For example, 2 is written as II in Roman numeral, just two ones added together. 12 is written as XII, which is simply X + II. The number 27 is written as XXVII, which is XX + V + II.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as IV. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as IX. There are six instances where subtraction is used:

- I can be placed before V (5) and X (10) to make 4 and 9.
- X can be placed before L (50) and C (100) to make 40 and 90.
- C can be placed before D (500) and M (1000) to make 400 and 900.

Write a C++ program that converts a given Roman numeral into its corresponding integer value.



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Examples:

Input: III : Output: 3
Input: LVIII : Output: 58
Input: MCMXCIV : Output: 1994